Python Activity-Based Learning
Program 3: Hospital Billing and Patient Classification

Problem Statement

Prepare a hospital bill by adding consultation, diagnostic, and treatment charges. Apply a concession according to patient category and classify the payment status.

Objectives

• Calculate the total hospital bill.

• Apply a concession based on the patient category.

• Validate payment and display the outstanding amount.

Assumptions

• Senior citizens receive a 10% concession; other patients receive no concession.

• A patient is classified as Fully Paid, Partially Paid, or Not Paid based on the amount paid.

• All charges and payment amounts must be zero or positive.

Algorithm

1. Read patient name and category.

2. Read consultation, diagnostic, and treatment charges.

3. Validate that charges are not negative.

4. Calculate the total bill.

5. Apply the category concession.

6. Read amount paid and calculate balance.

7. Classify payment status and print the bill.

Flowchart

```
START → Read patient details and charges → Validate charges → Calculate total → Check category → Apply concess
```

Pseudocode

```
BEGIN
  READ name, category, consultation, diagnosis, treatment
  IF any charge < 0 THEN DISPLAY invalid input and STOP
  total = consultation + diagnosis + treatment
  IF category is senior THEN concession = 10% of total
  ELSE concession = 0
  net_bill = total - concession
  READ paid
  IF paid < 0 OR paid > net_bill THEN DISPLAY invalid payment
  ELSE calculate balance and payment status
  DISPLAY bill
END
```

Python Source Code

```
name = input("Enter patient name: ")
category = input("Enter category (senior/other): ").lower()

consultation = float(input("Consultation charge: "))
diagnostic = float(input("Diagnostic charge: "))
treatment = float(input("Treatment charge: "))

if consultation < 0 or diagnostic < 0 or treatment < 0:
    print("Invalid charge. Charges cannot be negative.")
else:
    total = consultation + diagnostic + treatment
```

```
if category == "senior":
    concession = total * 0.10
else:
    concession = 0

net_bill = total - concession
print("Total charge: Rs.", round(total, 2))
print("Concession: Rs.", round(concession, 2))
print("Net bill: Rs.", round(net_bill, 2))

paid = float(input("Enter amount paid: "))

if paid < 0 or paid > net_bill:
    print("Invalid payment amount.")
else:
    balance = net_bill - paid
    if balance == 0:
        status = "Fully Paid"
    elif paid == 0:
        status = "Not Paid"
    else:
        status = "Partially Paid"

    print("Patient:", name)
    print("Payment status:", status)
    print("Balance due: Rs.", round(balance, 2))
```

Test Case

Patient: Meena; category: senior; consultation ■500; diagnostics ■1000; treatment ■2500; paid ■2000.
Expected: Total ■4000; concession ■400; net bill ■3600; balance ■1600; status Partially Paid.

Sample Output

```
Total charge: Rs. 4000.0
Concession: Rs. 400.0
Net bill: Rs. 3600.0
Patient: Meena
Payment status: Partially Paid
Balance due: Rs. 1600.0
```

Conclusion

The program calculates a hospital bill, applies the senior-citizen concession, validates payment, and reports the patient's payment classification.